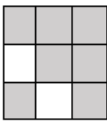
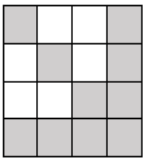
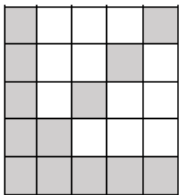
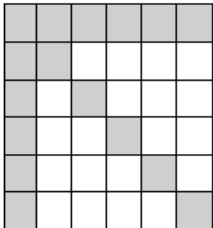


1	YTSS/2025/I/04												
	Show that $x = 2$ is a solution of $(k - 1)x^2 + 2x - 4k = 0$, where k is a constant, $k \neq 1$. [2]												
2	YTSS/2025/I/03												
	Solve the inequality $\frac{2 - x}{3} \leq \frac{1 + x}{2}$ [2]												
3	CCHY/2025/I/07												
	Rearrange the formula $\frac{x}{3} = \left(\frac{y}{7z - 6w}\right)^{\frac{1}{3}}$ to make z the subject. [3]												
4	CCHY/2025/I/15												
	Tom has 46 blue and 29 red balls. He wants to use all the balls to fill a ball pit. He wants at least 70% of the balls to be blue. Find the smallest number of extra blue balls he needs to add to his ball collection. [3]												
5	CCHY/2025/I/19												
	Given that n is an integer and $n > 1$, decide whether each statement in the table is true or false . For each statement write true or false in the table If you write false , give an example to justify your decision. [3]												
	<table><tr><th>Statement</th><th>True or False</th><th>Example (if false)</th></tr><tr><td>$n^3 > 1$</td><td></td><td></td></tr><tr><td>$\frac{1}{n} > \frac{1}{n^2}$</td><td></td><td></td></tr><tr><td>$(n - 1)(n + 3)$ is always odd</td><td></td><td></td></tr></table>	Statement	True or False	Example (if false)	$n^3 > 1$			$\frac{1}{n} > \frac{1}{n^2}$			$(n - 1)(n + 3)$ is always odd		
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6	CCHY/2025/I/26 In sequence A, the first 3 terms are $a \quad ar \quad ar^2$, where a and r are constants. In sequence B, the first term is a and the same non-zero number is added each time to obtain the next term. The first 3 terms of sequence A are equal to the first, third and sixth term of Sequence B respectively. (a) Show that $2r^2 - 5r + 3 = 0$. <i>Answer</i> [3] (b) Find the value of r, justifying your answer. [2]
7	CCHY/2025/II/08 StreamGuru is a legal online media services provider that allows users to download music albums over the Internet. It offers subscribers two choices of StreamGuru Plans, Standard and Elite. (a) Under the Standard Plan, subscribers can download x albums in one hour. Write an expression, in terms of x, for the time taken, in minutes, to download one album under the Standard Plan. [1] (b) Under the Elite Plan, subscribers can download 5 more albums in one hour compared to the Standard Plan. Write an expression, in terms of x, for the time taken, in minutes, to download one album under the Elite Plan. [1] (c) A StreamGuru subscriber can download an album under the Elite Plan 2 minutes faster than under the Standard Plan. Write an equation in x to represent this information and show that it reduces to $x^2 + 5x - 150 = 0$. [3] (d) Solve the equation $x^2 + 5x - 150 = 0$. [3] (e) How many minutes does it take for a Elite Plan subscriber to download 2 music albums? [2] (f) Find the percentage of time saved by downloading one music album under the Elite Plan compared to under the Standard Plan. [2]

8	SST/2025/II/02
(a)	Solve $7(x - 2) - 3(2x - 1) = 8(3x + 5) - 6$. [2]
(b)	Solve the inequality $-3x + 4 < -2x - 6$. [1]
(c)	It is given that $p = \frac{r^2q - 2}{q - r^2}.$
(i)	Find p when $r = \frac{1}{2}$ and $q = -1$. [1]
(ii)	Express r in terms of p and q . [3]
(d)	Solve the equation $\frac{x+3}{2x^2-8x+6} + \frac{5x}{3-x} = \frac{2}{x-1}$. [5]
9	SST/2025/II/03
	Two water taps, X and Y , together can fill a tank in 6 hours. Tap X takes x hours to fill the tank alone. Tap Y takes 25 hours less than the Tap X to fill the tank alone.
(a)	Write down an expression, in terms of x , for the fraction of the tank that Tap X will fill up in one hour . [1]
(b)	Write an expression, in terms of x , for the fraction of the tank that Tap Y will fill up in one hour. [1]
(c)	Write down an equation to represent the rate of filling the tank by the 2 taps and show that it reduces to $x^2 - 37x + 150 = 0$. [3]
(d)	Solve the equation $x^2 - 37x + 150 = 0$. [3]
(e)	Calculate how long it would take to fill the tank from empty using Tap Y Give your answer in hours and minutes, correct to the nearest minute . [2]

10	GMSS/2025/I/23
	Given that $5x = \frac{\sqrt{ny+3}}{2n}$, express y in terms of x and n. [3]
11	GMSS/2025/II/05
	A car has a fuel capacity of x litres (l). It can travel 800 km when it travels in the city. When it travelled the same distance on the highway, it used 10 litres of fuel lesser than it travelled in the city. (a) Find, in terms of x, the rate of fuel used, when the car travelled (i) in the city, [1] (ii) on the highway. [1] (b) The rate at which the fuel is used on the highway is 2 km/l more than that in the city. Write down an equation to represent this information and show that it reduces to $x^2 - 10x - 4000 = 0.$ <i>Answer</i> [3] (c) Solve the equation $x^2 - 10x - 4000 = 0$. Give your solutions correct to two decimal places. [3] (d) Determine the maximum distance the car can travel on the highway with x litres of fuel. [2]

12	ACSBR/2025/I/08 A packet of biscuit cost \$ a. A packet of drink cost \$ b. Scott spent \$ 11.80 on 4 packets of biscuits and 2 packets of drinks. Sierra spent \$ 11.10 on 3 packets of biscuits and 3 packets of drinks. Form and solve two simultaneous equations to find the cost of a packet of biscuit and a packet of drink. [3]
13	ACSBR/2025/II/07 The first 4 diagrams in the sequence are shown below.  Diagram 1  Diagram 2  Diagram 3  Diagram 4 (a) Find the number of grey boxes in Diagram 5. [1] (b) Find, in terms of n, an expression for the number of grey boxes in Diagram n. [2] (c) Diagram n has 210 white squares. Form an equation in terms of n to represent this information and show that it reduces to $n^2 + n - 210 = 0$. <i>Answer</i> [2] (d) (i) Solve the equation $n^2 + n - 210 = 0$. [3] (ii) Explain why one of the solutions to (d)(i) must be rejected. <i>Answer</i> [1]

Solution

1	When $x = 2$ ---[M1] for sub $x = 2$ LHS $= (k-1)(2)^2 + 2(2) - 4k$ $= 4k - 4 + 4 - 4k$ ---[A1] for expansion and show $=0$												
2	$\frac{2-x}{3} \leq \frac{1+x}{2}$ $2(2-x) \leq 3(1+x)$ ---[M1] $4-2x \leq 3+3x$ $-5x \leq -1$ $x \geq \frac{1}{5}$ ---[A1]												
3	$\frac{x^3}{27} = \frac{y}{7z-6w}$ $7x^3z - 6x^3w = 27y$ $7x^3z = 27y + 6x^3w$ $z = \frac{27y + 6x^3w}{7x^3}$												
4	$\frac{46+x}{46+29+x} \geq \frac{7}{10}$ $460+10x \geq 525+7x$ $3x \geq 65$ $x \geq 21.666$ Smallest $x = 22$												
5	<table><tr><th>Statement</th><th>True or False</th><th>Example (if false)</th></tr><tr><td>$n^3 > 1$</td><td>True</td><td></td></tr><tr><td>$\frac{1}{n} > \frac{1}{n^2}$</td><td>True</td><td></td></tr><tr><td>$(n-1)(n+3)$ is always odd</td><td>False</td><td>When $n = 3$, $(3-1)(3+3) = 12$, which is even</td></tr></table>	Statement	True or False	Example (if false)	$n^3 > 1$	True		$\frac{1}{n} > \frac{1}{n^2}$	True		$(n-1)(n+3)$ is always odd	False	When $n = 3$, $(3-1)(3+3) = 12$, which is even
Statement	True or False	Example (if false)											
$n^3 > 1$	True												
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$(n-1)(n+3)$ is always odd	False	When $n = 3$, $(3-1)(3+3) = 12$, which is even											

6

(a)

Let the same non-zero number be d . Then we have

$$ar = a + 2d$$

$$ar^2 = a + 5d$$

$$d = \frac{ar - a}{2}$$

$$d = \frac{ar^2 - a}{5}$$

Equating the two equations

$$\frac{ar - a}{2} = \frac{ar^2 - a}{5}$$

$$\frac{r - 1}{2} = \frac{r^2 - 1}{5}$$

$$5r - 5 = 2r^2 - 2$$

$$2r^2 - 5r + 5 - 2 = 0$$

$$2r^2 - 5r + 3 = 0$$

(b)

$$2r^2 - 5r + 3 = 0$$

$$(2r - 3)(r - 1) = 0$$

$$r = \frac{3}{2} \quad \text{or} \quad r = 1$$

r cannot be 1 because the difference between consecutive terms in sequence B is non-zero.

Thus, $r = \frac{3}{2}$.

7

(a)

$$\frac{60}{x}$$

(b)

$$\frac{60}{x+5}$$

(c)

$$\frac{60}{x} - \frac{60}{x+5} = 2$$

$$60(x+5) - 60x = 2x(x+5)$$

$$60x + 300 - 60x = 2x^2 + 10x$$

$$2x^2 + 10x - 300 = 0$$

$$x^2 + 5x - 150 = 0 \text{ (Shown)}$$

(d)

$$x^2 + 5x - 150 = 0$$

$$(x - 10)(x + 15) = 0$$

$$x = 10 \quad \text{or} \quad x = -15$$

(e)

Amount of time under Elite Plan for 2 music albums

$$= \frac{60}{10+5} \times 2$$

$$= 8 \text{ minutes}$$

(f)

$$\begin{aligned} \text{Percentage of time saved} &= \frac{6-4}{6} \times 100\% \\ &= 33\frac{1}{3}\% \end{aligned}$$

8

(a)

$$7x - 14 - 6x + 3 = 24x + 40 - 6$$

$$x = -\frac{45}{23} = -1\frac{22}{23}$$

Based on context,
Answer must be
exact and shown as
mixed number

(b)

$$-x < -10$$

$$x > 10$$

(ci)

$$p = \frac{\left(\frac{1}{2}\right)^2 (-1) - 2}{(-1) - \left(\frac{1}{2}\right)^2} = \frac{-9/4}{-5/4} = \frac{9}{5} = 1\frac{4}{5}$$

Or 1.8

(ci)

$$p(q - r^2) = r^2q - 2$$

$$pq - pr^2 = r^2q - 2$$

$$pr^2 + r^2q = pq + 2$$

$$r^2(p + q) = pq + 2$$

$$r^2 = \frac{pq + 2}{p + q}$$

$$r = \pm \sqrt{\frac{pq + 2}{p + q}}$$

(d)

$$\frac{x+3}{2(x-3)(x-1)} + \frac{5x}{3-x} = \frac{2}{x-1}$$
$$\frac{x+3}{2(x-3)(x-1)} - \frac{5x}{x-3} = \frac{2}{x-1}$$
$$\frac{x+3}{2(x-3)(x-1)} - \frac{5x(2)(x-1)}{2(x-3)(x-1)} = \frac{2(2)(x-3)}{2(x-3)(x-1)}$$
$$x+3-10x(x-1)=4(x-3)$$
$$10x^2-7x-15=0$$

$$x = \frac{-(-7) \pm \sqrt{(-7)^2 - 4(10)(-15)}}{2(10)}$$
$$x = 1.62 \quad \text{or} \quad -0.92$$
$$(2 \text{ d.p.}) \quad (2 \text{ d.p.})$$

9

(a)

$$\frac{1}{x}$$

(b)

$$\frac{1}{x-25}$$

(c)

$$\frac{1}{x} + \frac{1}{x-25} = \frac{1}{6}$$

Equation that represents
given context.

$$\frac{x+x-25}{x(x-25)} = \frac{1}{6}$$

$$6(2x-25) = x(x-25)$$

$$12x-150 = x^2-25x$$

$$x^2-25x-12x+150=0$$

$$x^2-37x+150=0 \text{ (shown)}$$

	(d) $x^2 - 37x + 150 = 0$ $x = \frac{-(-37) \pm \sqrt{(-37)^2 - 4(1)(150)}}{2(1)}$ $x = 32.365 \text{ or } x = 4.634$ $x = 32.4 \text{ or } x = 4.63 \text{ (3 s.f.)}$ (e) Reject $x = 4.63$ as the time taken, $x - 25$, cannot be negative. Time Tap Y takes to fill up the tank completely $= 32.365 - 25$ $= 7.365$ $= 7 \text{ hours } 22 \text{ minutes (nearest minute)}$ Must show substitution of values. Important to test both values and justify any rejection of solution.	
10	$5x = \frac{\sqrt{ny+3}}{2n}$ $10nx = \sqrt{ny+3}$ $100n^2x^2 = ny+3$ $y = \frac{100n^2x^2 - 3}{n}$	
11	5ai 5aia	$\frac{800}{x} \text{ km/l}$ $\frac{800}{x-10} \text{ km/l}$

	5b $\frac{800}{x-10} - \frac{800}{x} = 2$ $\frac{800x - 800x + 8000}{x^2 - 10x} = 2$ $4000 = x^2 - 10x$ $x^2 - 10x - 4000 = 0 \text{ (shown)}$
	5c $x^2 - 10x - 4000 = 0$ $x = \frac{-(-10) \pm \sqrt{(-10)^2 - 4(1)(-4000)}}{2(1)}$ $= \frac{10 \pm \sqrt{16100}}{2}$ $= 68.443 \text{ or } -58.443$ $= 68.44 \quad -58.44 \text{ (2 dp)}$
	5d Highway distance with $x = 68.44$ $= 68.44 \times \frac{800}{68.44 - 10}$ $= 936.877516$ Max distance = 936 km
12	$4a + 2b = 11.80$ $3a + 3b = 11.10$ $12a + 6b = 35.4$ $6a + 6b = 22.2$ $12a + 6b - (6a + 6b) = 35.4 - 22.2$ $a = 2.20$ $b = 1.50$

13	(a)	19
	(b)	$3n + 4$
	(c)	$(n + 2)^2 - (3n + 4) = 210$
		$n^2 + 4n + 4 - 3n - 4 = 210$
		$n^2 + n - 210 = 0$ (Shown)
	(di)	$n = \frac{-1 \pm \sqrt{1^2 - 4(1)(-210)}}{2(1)}$
		$n = 14$ or $n = -15$
	(dii)	$n = -15$ may be rejected as sequence/diagram number cannot take on a negative value.